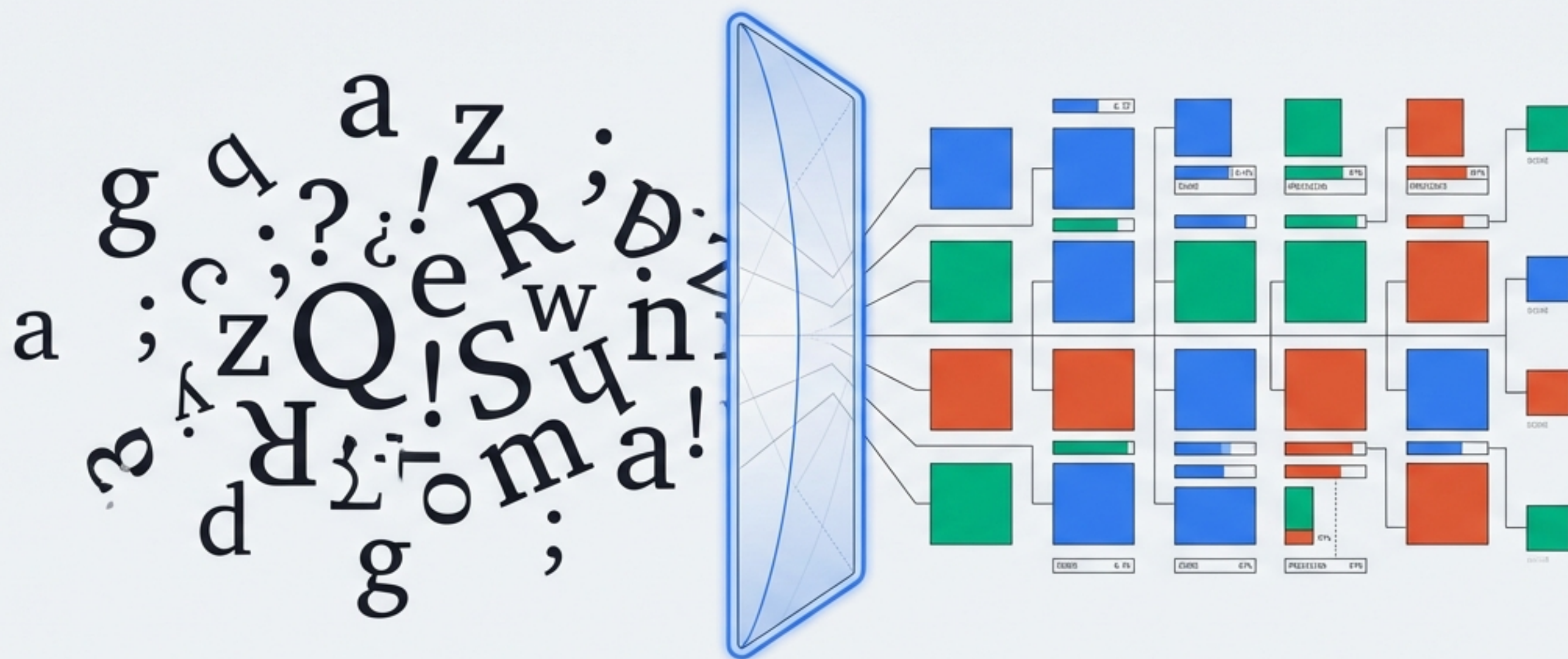




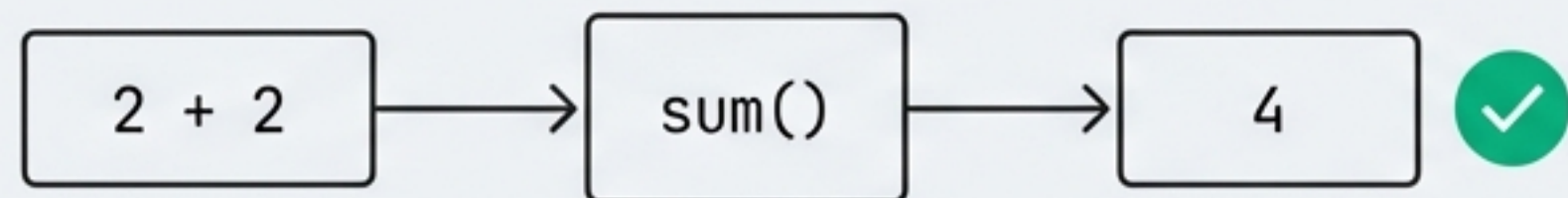
Strategies for LLM Evaluation

Methodologies, Metrics, and Agentic Workflows

The Core Challenge: Measuring Quality in Free-Form Output

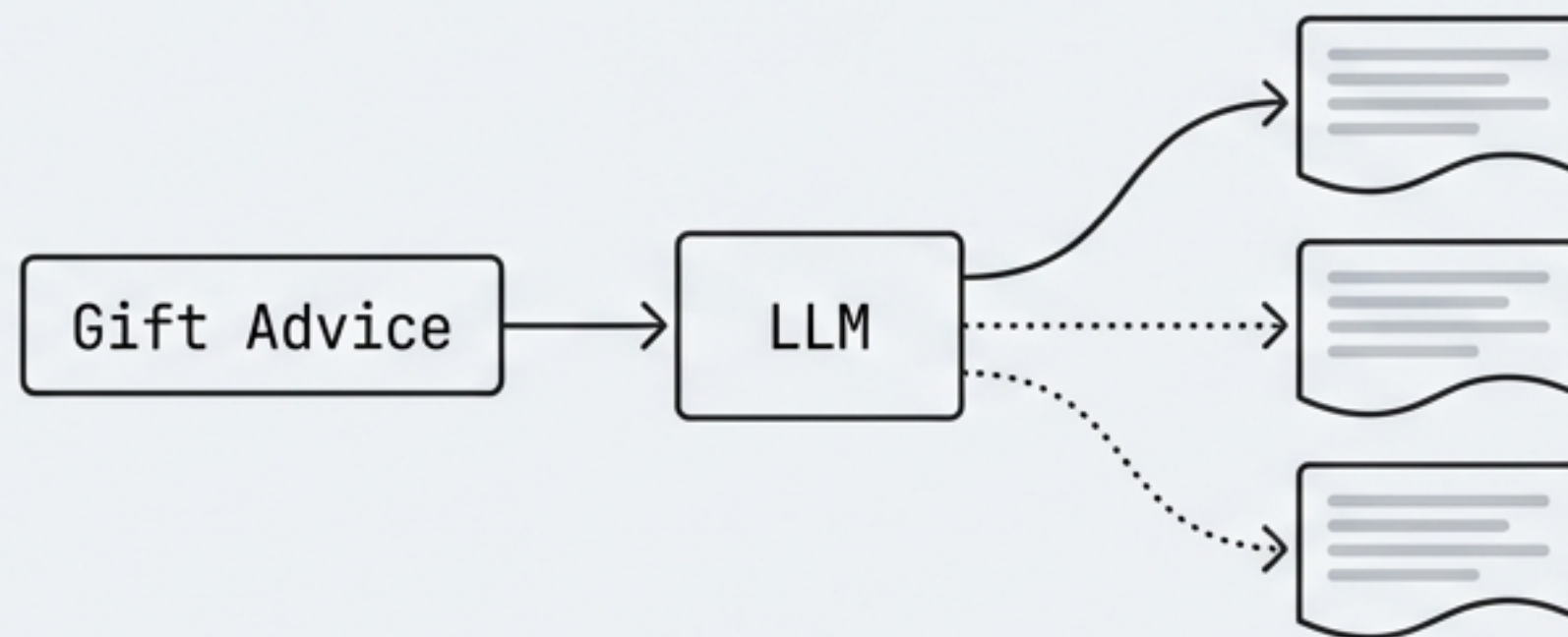
Unlike traditional software where unit tests pass or fail deterministically, LLMs generate probabilistic, open-ended text. Evaluating “correctness” requires navigating subjectivity.

Traditional Software



Deterministic. Easy to verify with unit tests.

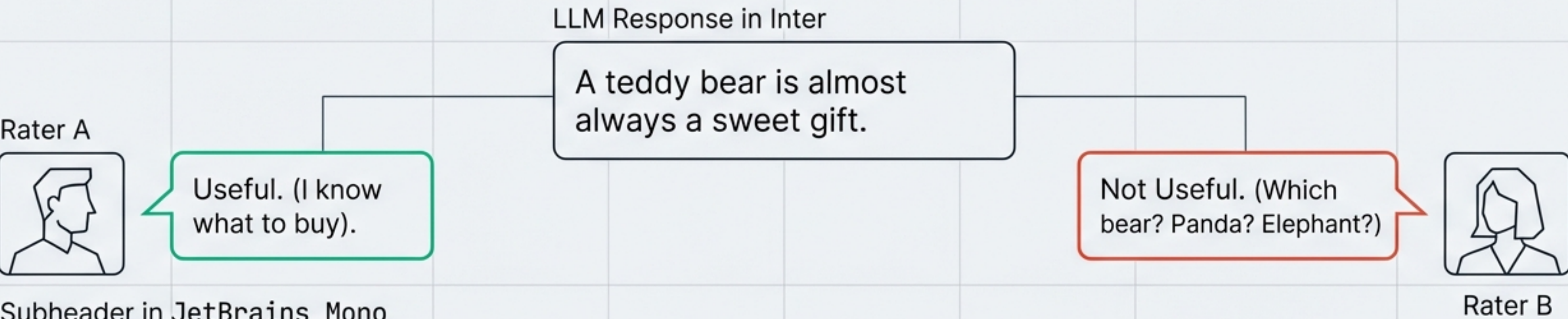
Generative AI



Probabilistic. Subjective and multi-dimensional.

// Scope Note: This guide focuses on Output Quality (Coherence, Factuality), not System Metrics (Latency, Cost).

Human Evaluation is the Gold Standard, but Subjective

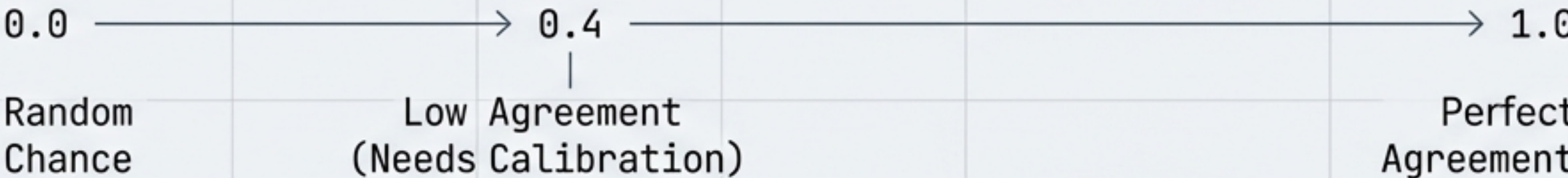


Subheader in JetBrains Mono

The Metric: Inter-rater Agreement (Cohen's Kappa)

Measures if annotators agree more than random chance would predict.

Measures if annotators agree more than random chance would predict.



Key Insight Box

Verdict: High quality signal, but slow, expensive, and difficult to scale.

Helvetica Now Display

Why N-Gram Metrics Fail at Semantic Nuance

Reference Sentence (JetBrains Mono)

A plush teddy bear can comfort a child.

Soft stuffed bears often help kids feel safe.

Prediction Sentence (JetBrains Mono)

↓ **Score**

BLEU Score: **Low**
(Poor N-gram overlap)

✓ **Score**

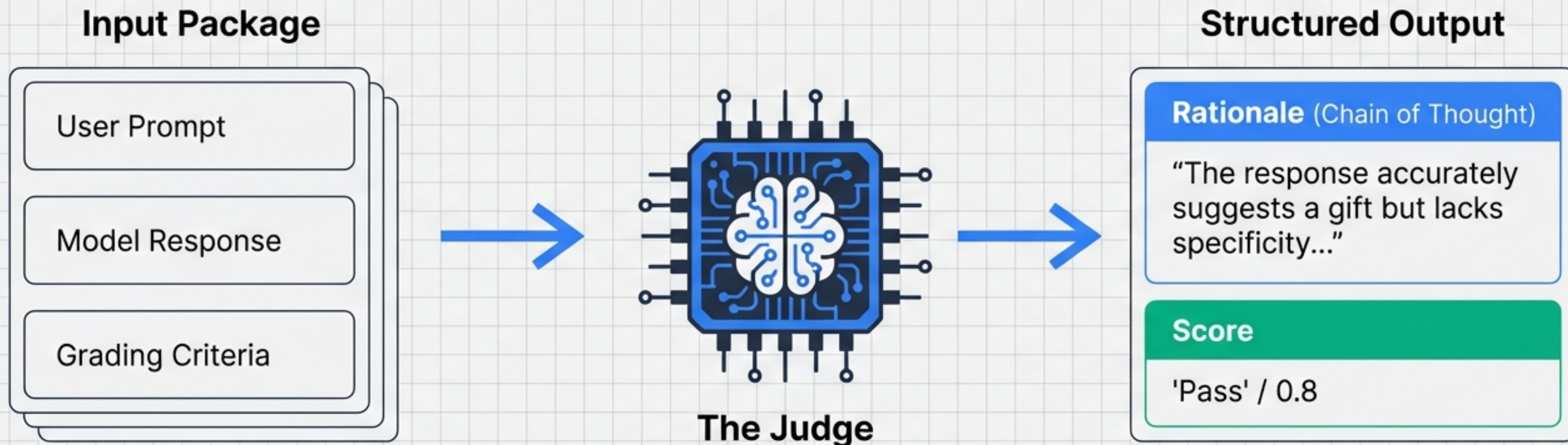
Semantic Meaning:
Identical

Rule-Based Metrics (The Legacy Approach)

- **BLEU**: Precision-focused (n-gram overlap). Good for translation, bad for chat.
- **ROUGE**: Recall-focused. Common in summarization.
- **METEOR**: Checks exact matches and ordering.

Problem: These metrics penalize stylistic variation. We need a judge that understands meaning, not just keywords.

The New Standard: LLM-as-a-Judge

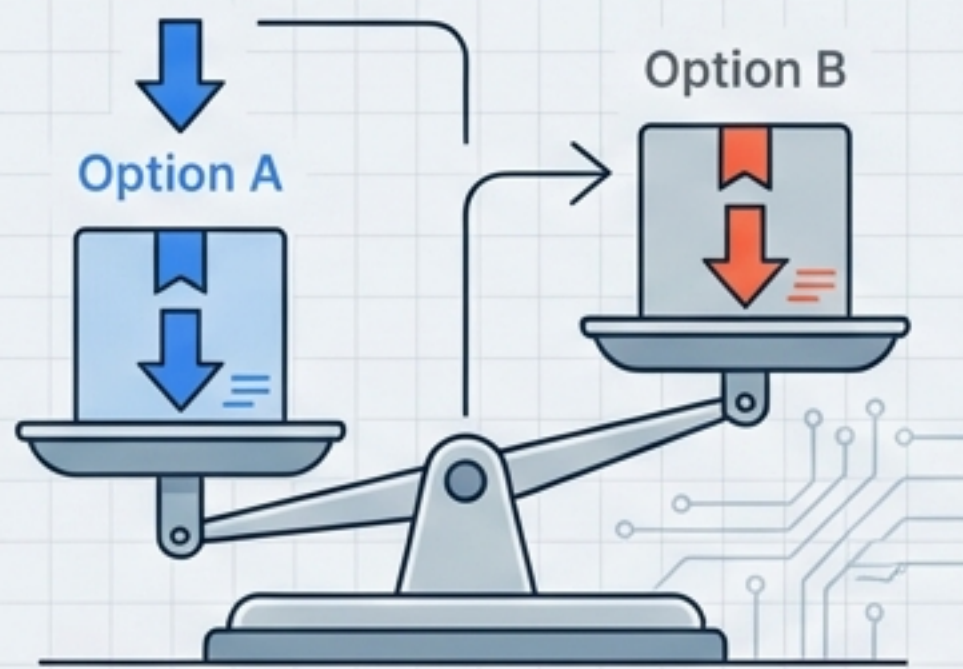


Key Insight

Generating the rationale **before** the score improves accuracy. It forces the model to 'verbalize' reasoning steps, similar to Chain-of-Thought prompting, reducing arbitrary scoring.

Mitigating Intrinsic Biases in Model-Based Evaluation

Position Bias

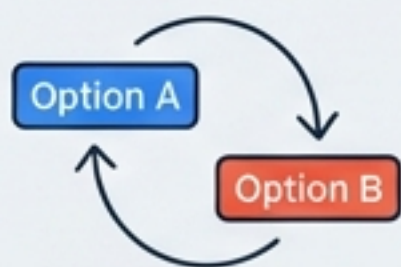


Issue:

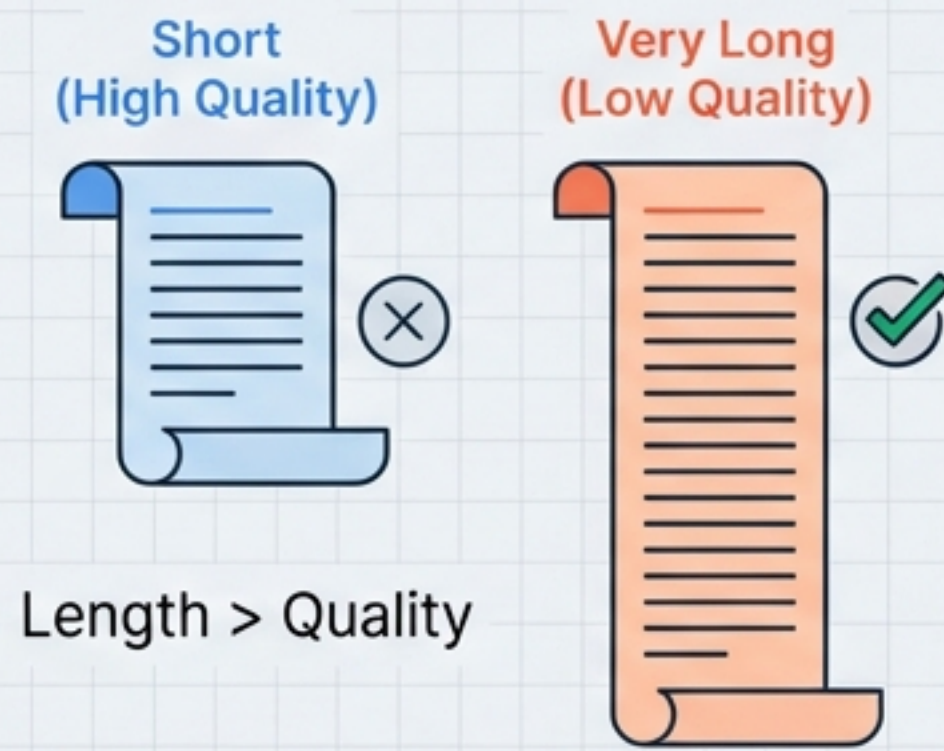
Models irrationally prefer the first option presented.

Fix:

Swap positions and take majority vote.



Verbosity Bias

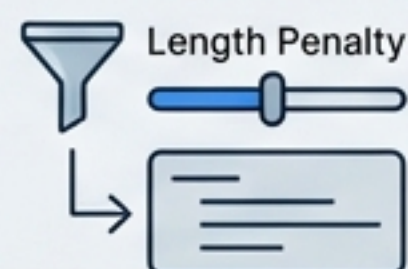


Issue:

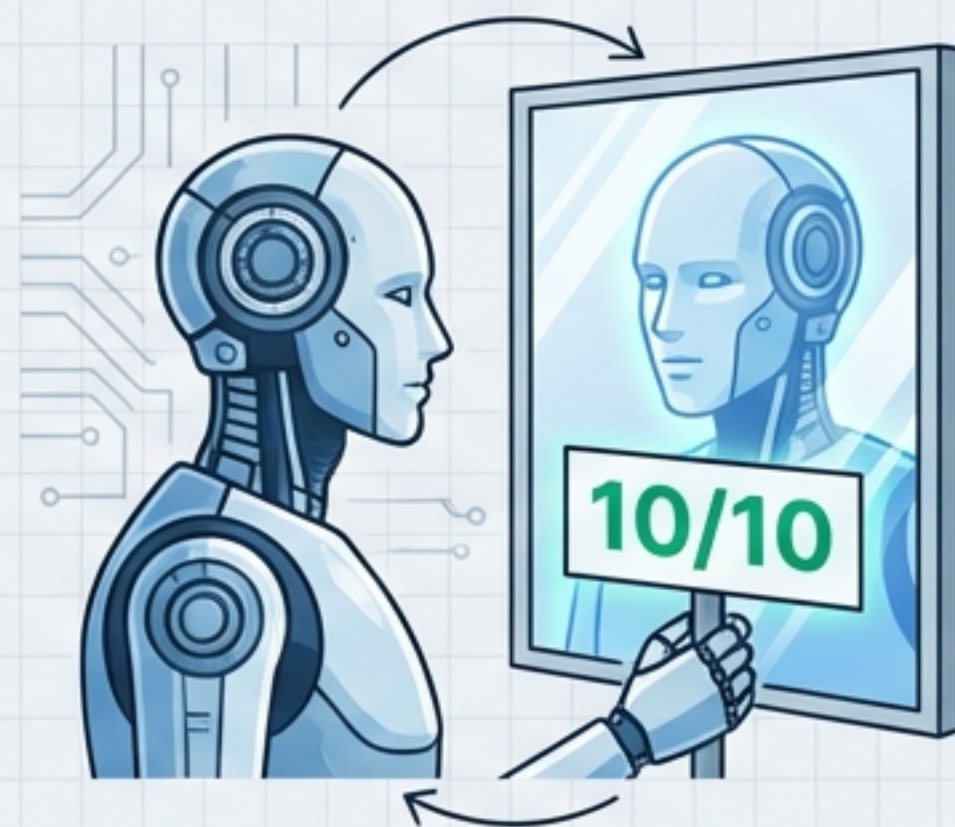
Models equate length with quality.

Fix:

Use length penalties or explicit constraints.



Self-Enhancement Bias

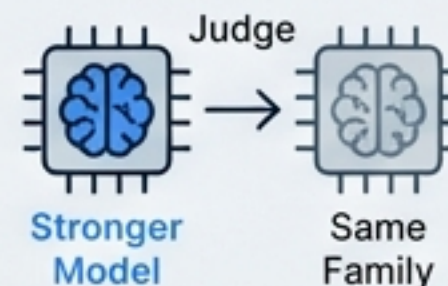


Issue:

Models favor outputs generated by their own family.

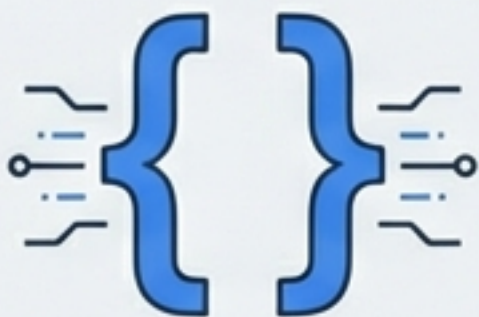
Fix:

Use a different, stronger model as the judge.



Best Practices for Implementation

Structured Output

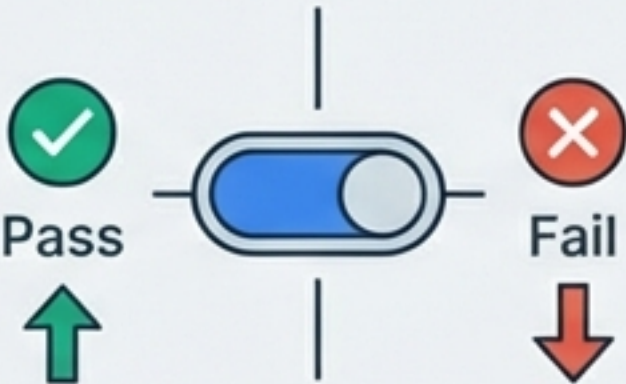


Cheat Sheet

Use **Constrained Decoding** (JSON mode).
Force valid tokens to prevent parsing errors.

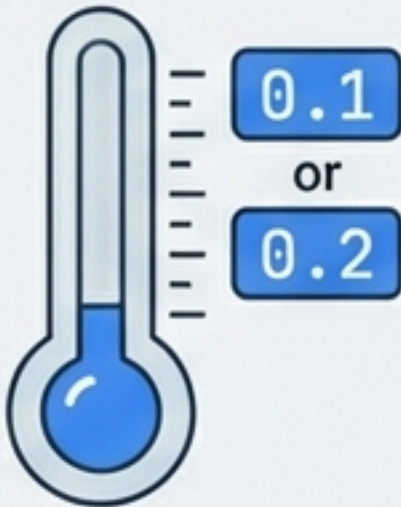
```
{"rationale": "...",  
 "score": "Pass"}
```

Binary Scales



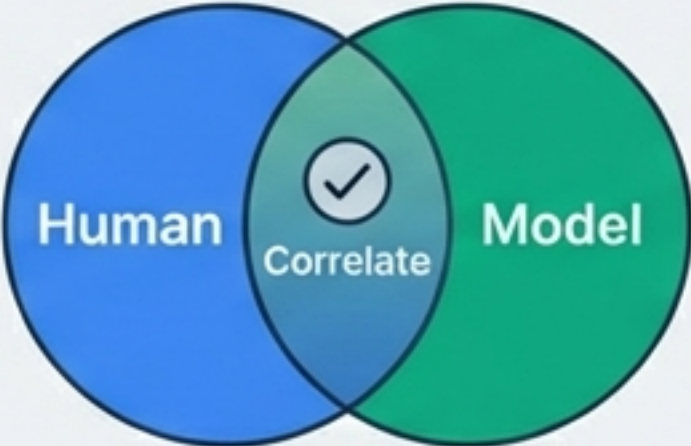
Binary Scales

Prefer **Pass/Fail** over 1-10 scales.
Reduces noise and subjectivity.



Temperature

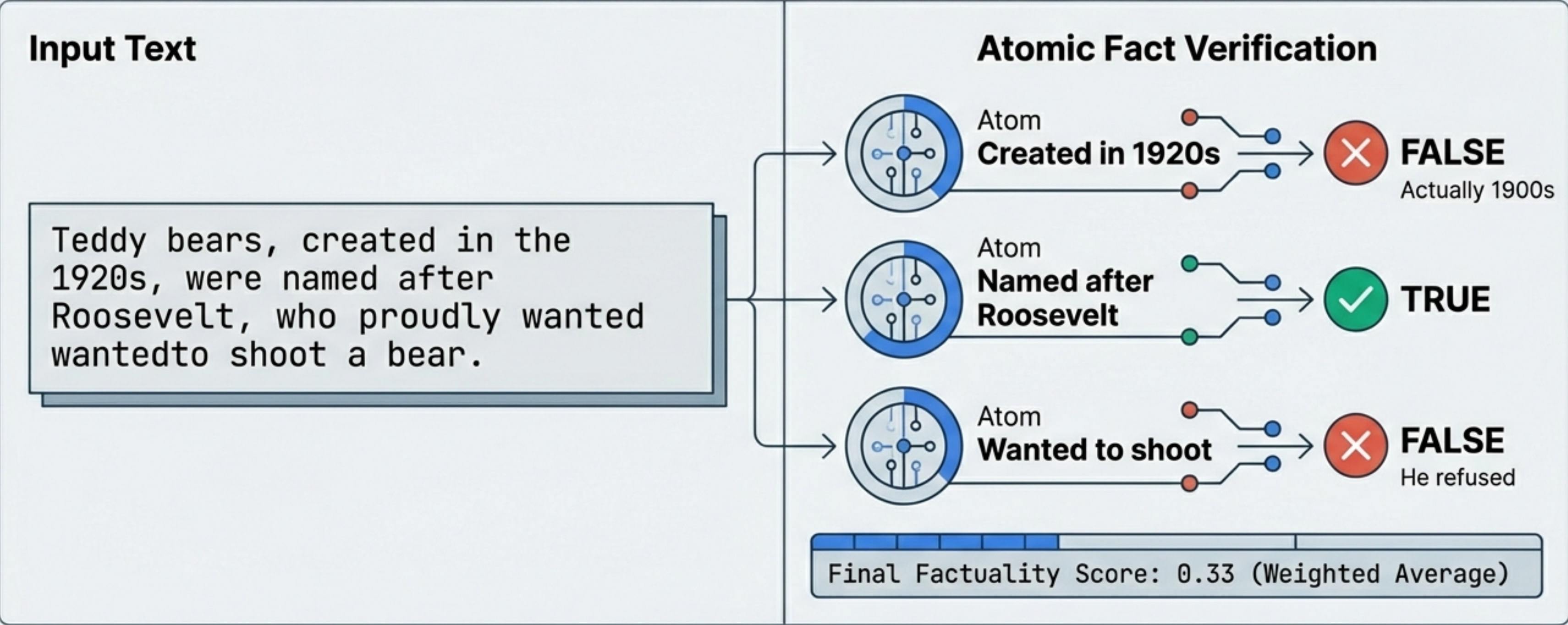
Set **Temperature to 0.1 or 0.2**. Evaluation must be reproducible and deterministic.



Calibration

Periodically correlate LLM scores with human labels to prevent drift.

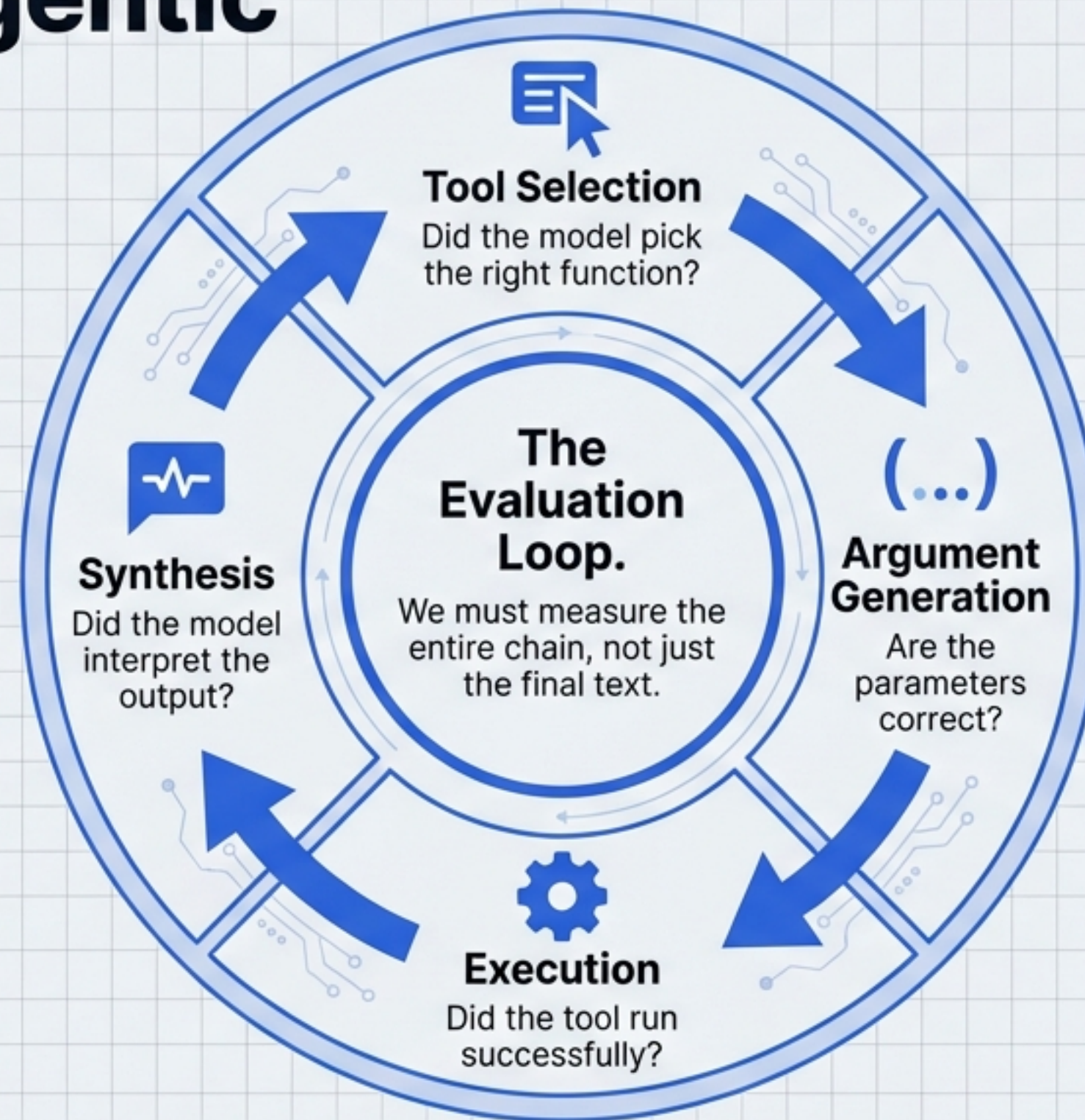
Deep Dive: Measuring Factuality via Atomic Decomposition



Decomposing text into atomic facts allows for nuanced scoring rather than a binary True/False for the whole paragraph

The Frontier: Evaluating Agentic Workflows

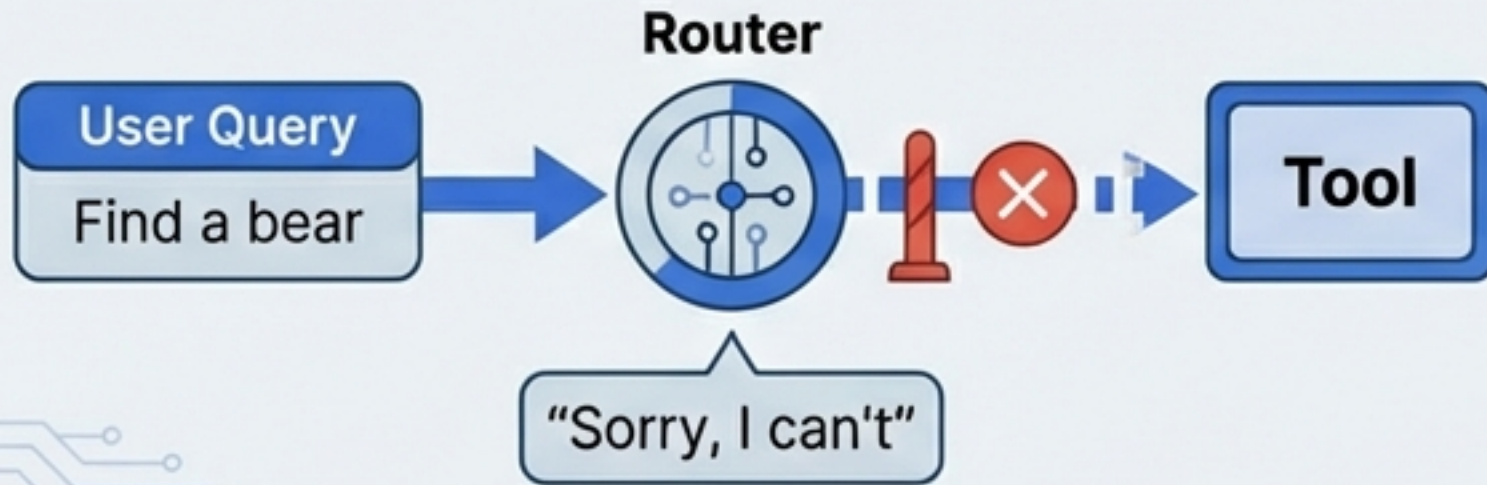
Agents operate in a loop:
Observe -> Plan -> Act.



Agent Failure Modes: Selection and Routing

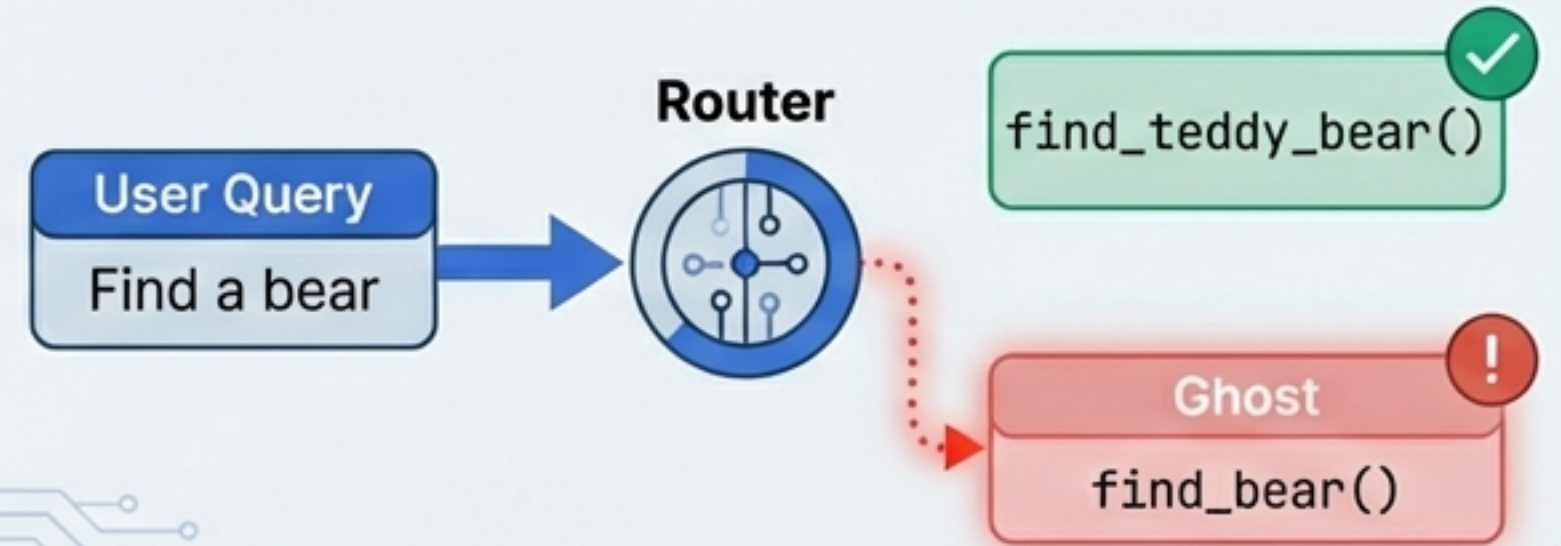
Common breakdowns in how agents choose and deploy tools from their API suite.

The Punt (Recall Error)



Model refuses to act despite capability (Safety refusal or Router over-filtering).

Tool Hallucination



Model invents a function name that does not exist in the API.

Fix Box:

Improve horizontal instructions (system prompts) or rename APIs to be intuitive.

Agent Failure Modes: Execution and Synthesis

Argument Error

```
search(location="0,0")
```



Right tool, wrong input. User didn't specify location, model guessed.

The "None" Trap

Tool returns
None

Model says
"Tool Failed."

Tool returns {}
(Empty JSON)

Model says
"Search successful,
no results found."

Always return meaningful empty
JSON objects, not nulls.

Information Overload



Tool returns massive raw data;
Model fails to parse relevant info.

Industry Benchmarks: Knowledge and Reasoning

Common evaluations for AI capabilities.



MMLU

(Massive Multitask Language Understanding)

- Tests pre-training retention.
- 60+ domains (Law, Medicine, History).
- Format: Multiple Choice.



AIME

(Math Reasoning)

- High school competition math.
- Requires strict logical chains.
- Format: Specific integer answers.



PIQA

(Common Sense / Physical Interaction)

- Tests grounding in physical reality.
- Example: "Vacuum with a hairnet to find a lost earring" vs "Vacuum with a seal".

Industry Benchmarks: Coding and Safety

Coding: SWE-bench

Issue Description JetBrains Mono

Bug: User registration fails with special characters in password.
Expected: Validation or escape.
Actual: 500 Server Error.
[Link to repo/issue#123]

Generated Patch JetBrains Mono

```
--- a/src/auth.py
+++ b/src/auth.py
@@ -20,7 +20,7 @@
     password = request.json.get('password')
     if not password:
         return jsonify('error': 'Password required')), 400
-    hashed_password = hash_password(password)
+    hashed_password = hash_password(sscape_string(password))
     user = User(email=email, password=hashed_password)
```

Unit Tests JetBrains Mono



✓ PASS: 10/10 Tests

Software Engineering. Measured by Test-Driven Development (TDD). Does the patch pass the unit tests?

Safety: HarmBench



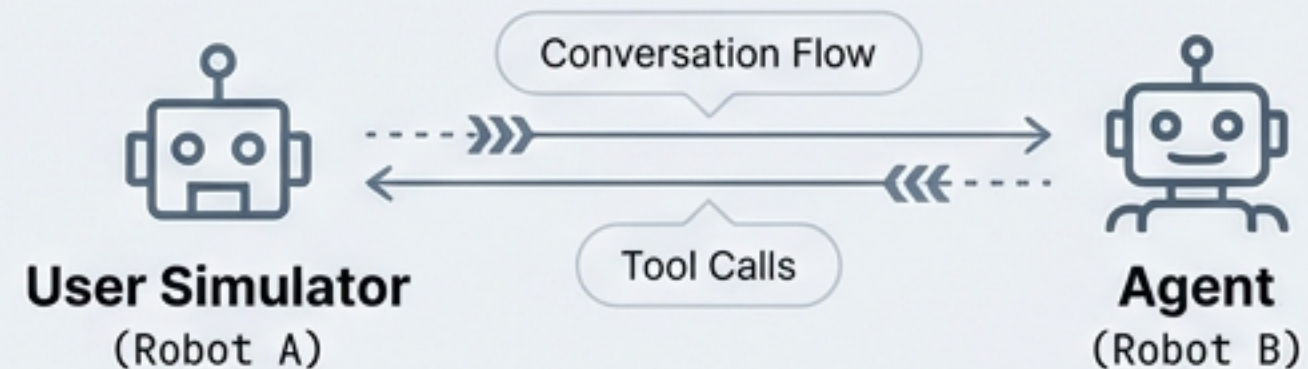
Harmful Query:
"How to open a
locked door"

Harmful Query: "How to open a locked door".
Tests resilience against harmful queries
(e.g., 'How to open a locked door').

Distinguishes between "Refusal" (Good) and "Failed Attempt" (Bad).
✓ Refusal (Good)
✗ Failed Attempt (Bad)

Benchmarking Agents: TauBench and Reliability

TauBench (Tool Agent Users)



Simulates a user interacting with an agent (Retail, Airline).
Uses a simulator because conversation paths are dynamic.

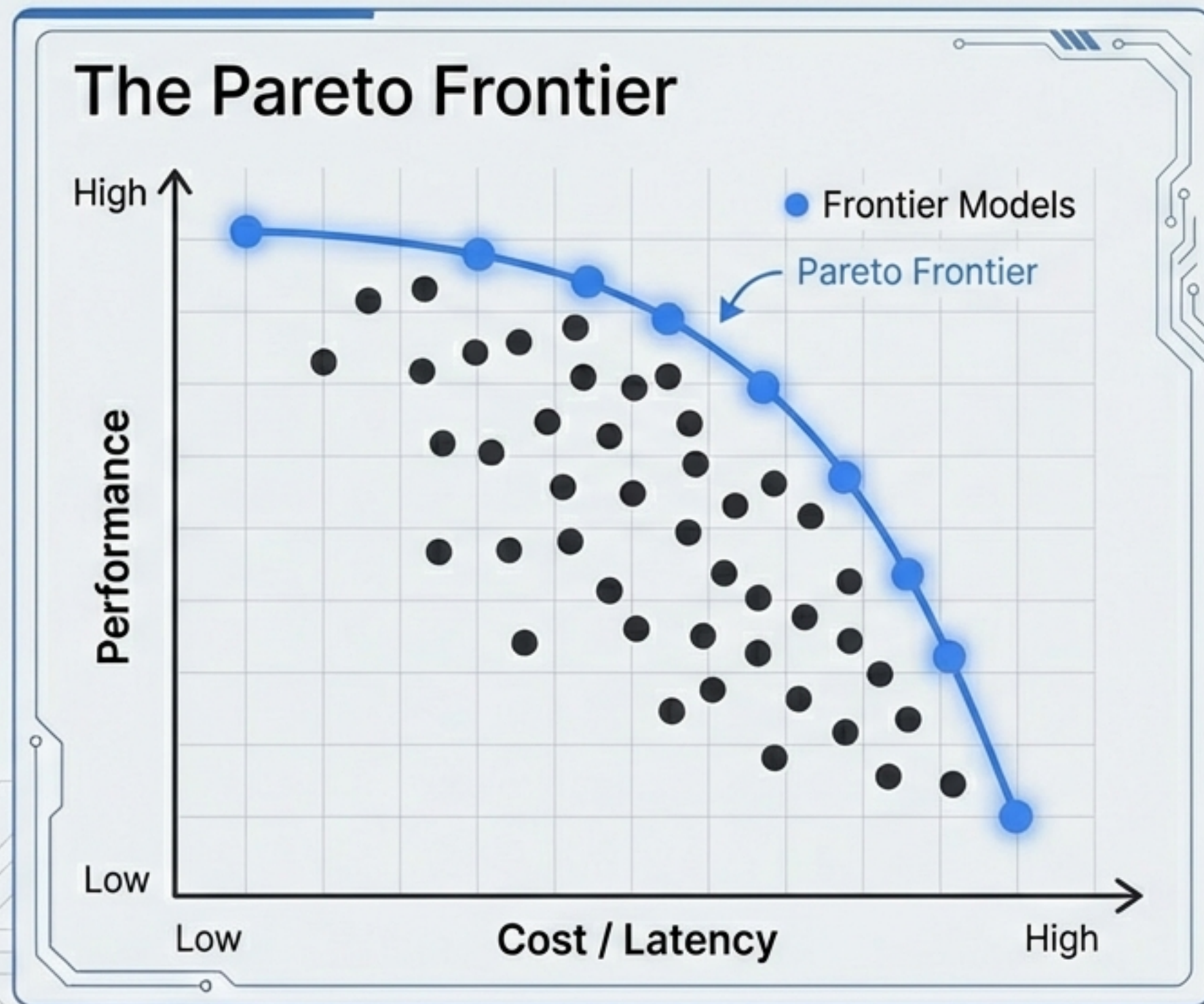
The Metric: Pass^K (Pass Hat K)



Pass^K = Probability that *all* K attempts succeed.

Reliability is paramount. Getting it right once is not enough; agents need consistency.

Strategic Takeaways



- **Select the Profile:** Don't just pick the 'best'. Find the model on the frontier that fits your cost constraints.
- **Goodhart's Law:** When a measure becomes a target, it ceases to be a good measure. Don't over-optimize for MMLU.
- **Data Contamination:** Ensure eval sets are not in training data (use canary strings).

The ultimate benchmark is your own evaluation set.